

QUESTIONS

Q1 
Industrial Production Control
Applet
25 marks

Q2 
Banking Transaction
Management Applet
25 marks

Q3 
Hospital Emergency
Management Applet
25 marks

Q4 
Smart Factory Monitoring and
Alert Applet
25 marks

4/4 answered

Q1 Industrial Production Control Applet

25 marks

1. Develop a Java applet for a manufacturing plant that monitors multiple machines simultaneously. Provide controls to Start, Stop, Pause, and Reset each machine. Use the Delegation Event Model with appropriate event classes and listener interfaces. Design the interface using suitable AWT layout managers, and provide a MenuBar for machine selection, monitoring, and report generation. The applet must dynamically update machine status and prevent invalid operations such as starting an already running machine.

Your Code

```
1 import java.awt.event.*;
2 import java.util.Scanner;
3
4 public class BankingApp implements ActionListener {
5
6     double balance = 5000.0;
7
8     public void actionPerformed(ActionEvent e) {
9         System.out.println("Event: " + e.getActionCommand());
10    }
11
12    public static void main(String[] args) {
13
14        Scanner sc = new Scanner(System.in);
15        BankingApp bank = new BankingApp();
16
17        while (true) {
18
19            System.out.println("\n===== BANKING SYSTEM =====");
20            System.out.println("1. Deposit");
21            System.out.println("2. Withdrawal");
22            System.out.println("3. Fund Transfer");
23            System.out.println("4. Balance Inquiry");
24            System.out.println("5. Exit");
25            System.out.print("Enter choice: ");
26
27            if (!sc.hasNextLine()) {
28                System.out.println("No input provided.");
```

Input

stdin input

Output

```
===== BANKING SYSTEM =====
1. Deposit
2. Withdrawal
3. Fund Transfer
4. Balance Inquiry
5. Exit
```

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Q2 Banking Transaction Management Applet

25 marks

Develop an interactive banking applet supporting Deposit, Withdrawal, Fund Transfer, and Balance Inquiry. Use AWT controls and menus to accept transaction details. Implement event handling with appropriate listener interfaces and validate conditions such as insufficient balance, invalid amount, and incomplete input. Use different layout managers to organize the transaction panels and display transaction status dynamically.

Your Code

```
1 import java.awt.*;  
2 import java.awt.event.*;  
3  
4 public class ManufacturingPlantMonitor extends Frame implements ActionListener {  
5  
6     Button[] startButtons = new Button[3];  
7     Button[] stopButtons = new Button[3];  
8     Button[] pauseButtons = new Button[3];  
9     Button[] resetButtons = new Button[3];  
10  
11     Label[] statusLabels = new Label[3];  
12  
13     String[] machines = {"Machine 1", "Machine 2", "Machine 3"};  
14     String[] status = {"Stopped", "Stopped", "Stopped"};  
15  
16     TextArea reportArea;  
17  
18     public ManufacturingPlantMonitor() {  
19  
20         setTitle("Manufacturing Plant Machine Monitor");  
21         setSize(700, 500);  
22         setLayout(new BorderLayout());  
23  
24         // Menu Bar  
25         MenuBar menuBar = new MenuBar();  
26  
27         Menu machineMenu = new Menu("Machine");  
28         MenuItem selectMachine = new MenuItem("Select Machine");
```

Input

stdin input

Output

```
Exception in thread "main"  
java.awt.HeadlessException  
at  
java.desktop/java.awt.GraphicsEnvironment.checkHeadless(GraphicsEnvironment.java:166)
```

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Q3 Hospital Emergency Management Applet

25 marks

Design a hospital emergency-management applet that accepts patient registration, emergency category, doctor selection, and treatment priority. Use Choice, Checkbox, TextField, TextArea, Button, and menus. Implement event handling to automatically assign priority and display alerts for critical patients. The system must prevent duplicate patient registration and validate all required inputs before confirmation.

Your Code

```

1 import java.awt.event.*;
2 import java.util.Scanner;
3
4 public class HospitalEmergency implements ActionListener {
5
6     String[] patients = new String[100];
7     int patientCount = 0;
8
9     String patientName;
10    String category;
11    String doctor;
12    String priority;
13
14    public void actionPerformed(ActionEvent e) {
15        System.out.println("Event: " + e.getActionCommand());
16    }
17
18    public static void main(String[] args) {
19
20        Scanner sc = new Scanner(System.in);
21        HospitalEmergency hospital = new HospitalEmergency();
22
23        System.out.println("===== HOSPITAL EMERGENCY SYSTEM =====");
24        System.out.print("Enter patient name: ");
25

```

Input

Ravi
1
2

Output

```

===== HOSPITAL EMERGENCY SYSTEM
=====
Enter patient name:
Emergency Category:
1. Critical
2. Serious

```

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4/4 answered

Q4 Smart Factory Monitoring and Alert Applet

25 marks

4. Develop a Java applet for monitoring a smart factory containing temperature, pressure, machine status, and production-level controls. Provide Normal, Warning, and Emergency modes using AWT components. Implement multiple event listeners to respond to user actions and generate appropriate alerts. Use BorderLayout, GridLayout, CardLayout, and menus to create different monitoring sections. The applet should dynamically switch between views and prevent conflicting machine operations.

Your Code

```
1 import java.awt.*;  
2 import java.awt.event.*;  
3  
4 public class SmartFactory implements ActionListener {  
5  
6     int temperature = 25;  
7     int pressure = 50;  
8     int production = 50;  
9     String machineStatus = "STOPPED";  
10    String mode = "NORMAL";  
11  
12    public void actionPerformed(ActionEvent e) {  
13  
14        String command = e.getActionCommand();  
15  
16        if (command.equals("NORMAL")) {  
17            mode = "NORMAL";  
18            System.out.println("Mode changed to NORMAL");  
19        }  
20  
21        else if (command.equals("WARNING")) {  
22            mode = "WARNING";  
23            System.out.println("Mode changed to WARNING");  
24        }  
25  
26        else if (command.equals("EMERGENCY")) {  
27            mode = "EMERGENCY";  
28            System.out.println("Mode changed to EMERGENCY");  
29        }  
30    }  
31 }
```

Input

stdin input

Output

```
Temperature: 25  
Pressure: 50  
Production Level: 50  
Machine Status: STOPPED  
Mode: NORMAL
```